

CRANBROOK

MATHEMATICS EXTENSION 2

2008

HIGHER SCHOOL CERTIFICATE

TRIAL EXAMINATION

General Instructions

- Reading time – 5 minutes
- Writing time – 3 hours
- All eight questions should be attempted
- Total marks available - 120
- All questions are worth 15 marks
- An approved calculator may be used
- All relevant working should be shown for each question
- Answer each question in a separate 8 page booklet.
- Standard integrals sheet at back of examination.

Question 1 (15 marks)

Marked by SKB

Marks

(a) Find $\int x \tan x^2 dx$.

2

(b) Use the substitution $u = \sqrt{x}$ to evaluate $\int_4^9 \frac{x}{\sqrt{x}(1+x)} dx$.

3

(c) Use the completion of squares method to find $\int \frac{-2}{\sqrt{3+2x-x^2}} dx$.

2

(d) (i) Find the real numbers a , b and c such that

2

$$\frac{2x^2 + 2x + 5}{(x^2 + 2)(1-x)} = \frac{ax + b}{x^2 + 2} + \frac{c}{1-x}.$$

(ii) Hence find $\int \frac{2x^2 + 2x + 5}{(x^2 + 2)(1-x)} dx$.

2

(e) Use integration by parts to evaluate $\int_1^5 \frac{\ln x}{\sqrt{x}} dx$.

4

Question 2 (15 marks) Marked by SKB **Marks**

(a) Evaluate $\int_{-1}^1 \frac{\tan^{-1} x}{1+x^4} dx$ 2

(b) Evaluate $\int_0^1 \sqrt{4-x^2} dx$ 4

(c) By using the result $\int_0^a f(x) dx = \int_0^a f(a-x) dx$
 evaluate $\int_0^{2\pi} \frac{x \cos x}{1+\sin^2 x} dx$ 3

(d) Let $I_n = \int_0^1 x(x^2-1)^n dx$ for $n = 0, 1, 2, \dots$

(i) Use integration by parts to show that 3

$$I_n = \frac{-n}{n+1} I_{n-1} \text{ for } n \geq 1.$$

(ii) Hence or otherwise show that 2

$$I_n = \frac{(-1)^n}{2(n+1)} \text{ for } n \geq 0.$$

(iii) Explain why $I_{2n} > I_{2n+1}$ for $n \geq 0$ 1

Question 3 (15 marks)

Marked by JSH

Marks

- (a) Let $z = 3 - i$ and $w = 2 + 4i$.
Find the following in the form $x + yi$.

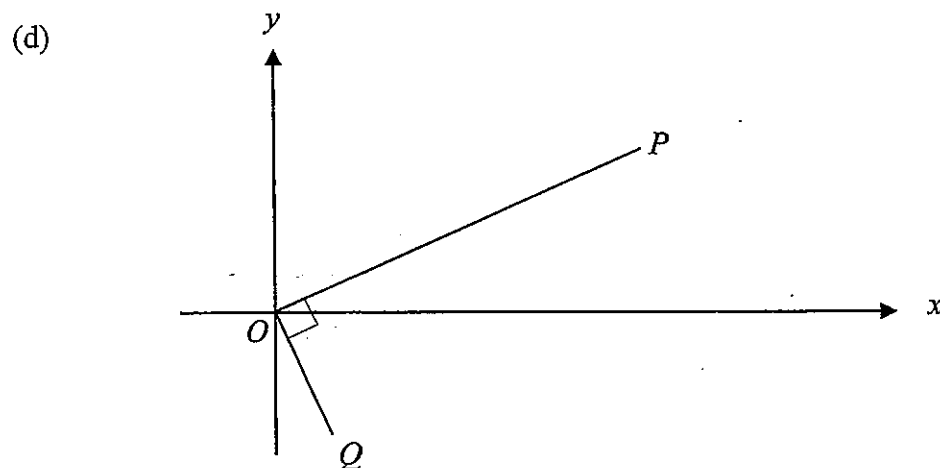
(i) $z\bar{w}$ 1

(ii) $\frac{z}{w}$ 1

- (b) (i) Express $1 + i$ in modulus-argument form. 2

- (ii) Hence, find the values of n , for which 3
 $(1 + i)^n + (1 - i)^n = 0$
 where n is a positive integer.

- (c) Sketch the region in the Argand diagram where the inequalities 2
 $|z - 1| \leq 1$ and $\frac{\pi}{4} \leq \arg(z - 1) \leq \frac{\pi}{2}$ both hold.



In the Argand diagram above, point P corresponds to the complex number z . 1

The triangle OPQ is a right-angled triangle and $OP = 3OQ$.
What is the complex number that corresponds to point Q ?

- (e) (i) Find all the solutions to the equation $z^6 = 1$ in the form $x + yi$. 2

- (ii) If ω is a non-real solution to the equation $z^6 = 1$, 2
 show that $\omega^4 + \omega^2 = -1$.

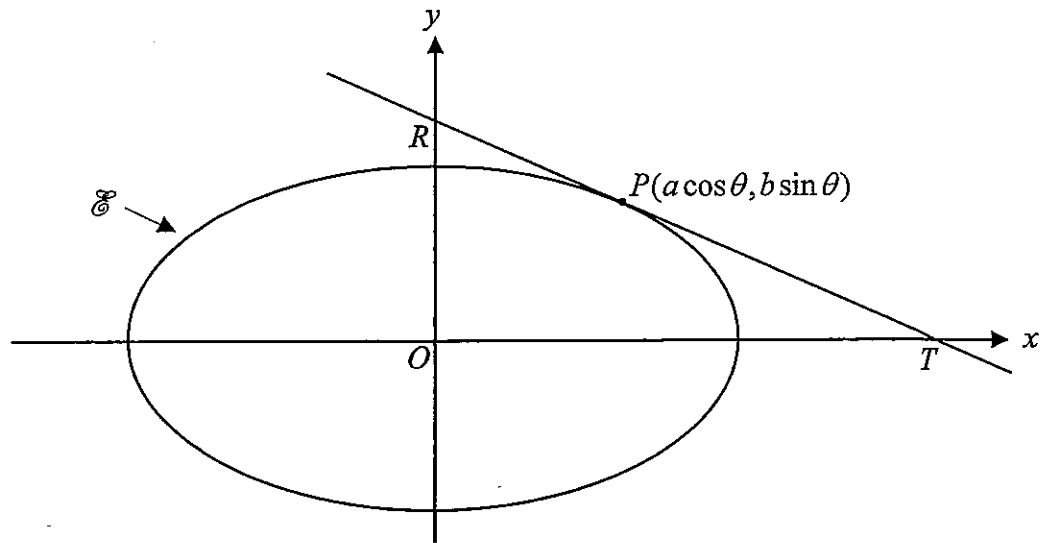
- (iii) By choosing one particular value of ω , explain with 1
 the aid of a diagram why $\omega^4 + \omega^2 = -1$.

Question 4 (15 marks)

Marked by JSH

Marks

(a)



The ellipse E with equation $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ shown in the diagram above, has a tangent at the point $P(a \cos \theta, b \sin \theta)$. The tangent cuts the x -axis at T and the y -axis at R .

- (i) Show that the equation of the tangent at the point P is 2
- $$\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1.$$

- (ii) If T is the point of intersection between the tangent at point P and one of the directrices of the ellipse, show that $\cos \theta = e$. 3

- (iii) Hence find the angle that the focal chord through P makes with the x -axis. 1

- (iv) Using similar triangles or otherwise, show that $RP = e^2 RT$. 3

- (b) $P(ct, \frac{c}{t})$, $t \neq 1$ lies on the hyperbola $xy = c^2$. The tangent and normal at P meet the line $y = x$ at T and N respectively. If O is the origin show that $OT \cdot ON = 4c^2$. Include a labelled diagram with your answer. 6

Question 5 (15 marks)

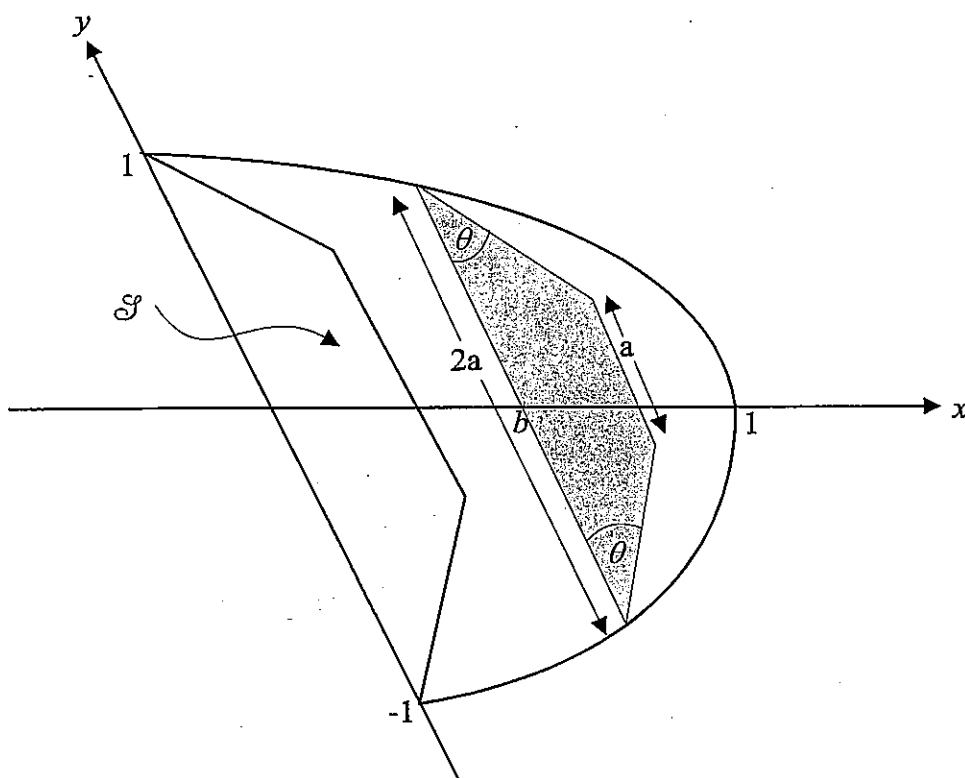
Marked by SKB

Marks

- (a) The region bounded by the curve $x = y^2$ and the line $x = 4$ is rotated about the line $y = 2$. Find the volume generated when:

- (i) Slices of thickness Δx are taken perpendicular to the x -axis in this region to create hollow cylindrical discs. 4
- (ii) Slices of thickness Δy are taken perpendicular to the y -axis in this region to create thin cylindrical shells. 4

(b)



A solid S has a semi-circular base in the x - y plane with its diameter along the y -axis.

Each cross-section of the solid running perpendicular to the x - y plane is a regular trapezium with its base sidelength twice that of its parallel sidelength. The angle between the base sidelength and the sides of the trapezium is θ .

A typical cross-section taken at $x = b$ is shown in the diagram.

- (i) Show that if $\theta = 45^\circ$, the area of the trapezium at $x = b$ is $\frac{3a^2}{4}$. 1

- (ii) Find the volume of the solid S when $\theta = 45^\circ$. 2
- (iii) Find the volume of the solid, $\mathcal{D}$, generated when the semi-circle is rotated through an angle of 90° about the y -axis. 1
- (iv) Find the values of θ for which the volume of S found in part (ii) is greater than the volume of $\mathcal{D}$. 3

Question 6 (15 marks) Marked by JSH **Marks**

(a) If $1-i$ is a zero of $P(x) = x^3 + ax^2 + bx + 6$, where $a, b \in \text{Real}$

(i) Evaluate a and b **4**

(ii) Hence fully factorise $P(x)$ over the complex field. **1**

(b) (i) Use De Moivre's theorem to express $\tan 5\theta$ in terms of powers of $\tan \theta$. **3**

(ii) Hence show $x^5 - 5x^4 - 10x^3 + 10x^2 + 5x - 1 = 0$ has roots **2**
 $1, \tan \frac{\pi}{20}, \tan \frac{9\pi}{20}, -\tan \frac{3\pi}{20}$ and $-\tan \frac{7\pi}{20}$.

(iii) By solving $x^5 - 5x^4 - 10x^3 + 10x^2 + 5x - 1 = 0$ another way, show that **5**

$$\tan \frac{9\pi}{20} + \tan \frac{\pi}{20} = 2 + 2\sqrt{5} \text{ and } \tan \frac{7\pi}{20} + \tan \frac{3\pi}{20} = 2\sqrt{5} - 2.$$

Question 7 (15 marks)

Marked by JSH

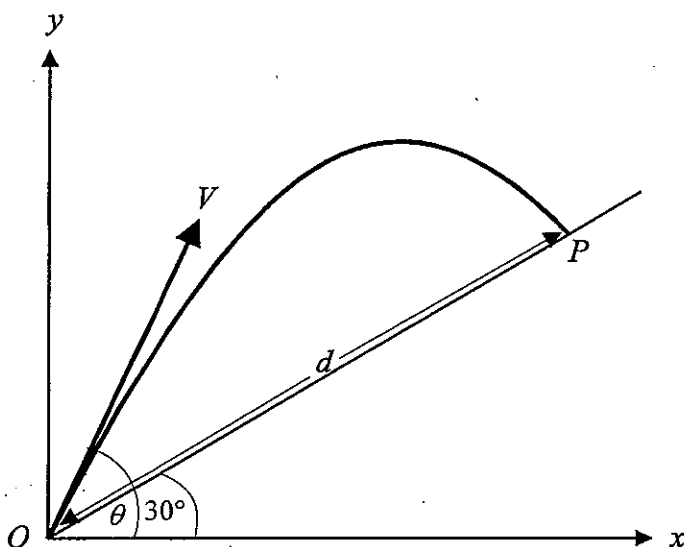
Marks

- (a) AB is a chord of a circle. X is a point on AB produced. XT is a tangent from X to the circle.

(i) Prove that $\triangle XAT$ is similar to $\triangle XTB$. 2

(ii) Deduce that $XT^2 = XA.XB$ 2

(b)



The diagram above shows the path of a particle which has been projected from point O at an angle of θ to the horizontal. The speed at which the particle was projected was $\sqrt{g}$ m/sec where g is the acceleration due to gravity. The particle lands at point P which lies on a plane inclined at an angle of 30° to the horizontal. The base of this inclined plane is at O and point P lies d metres from O . The position of the particle at time t seconds is given by

$$x = \sqrt{g} t \cos \theta$$

$$\text{and } y = \sqrt{g} t \sin \theta - \frac{1}{2} g t^2$$

- (i) Show that the path of trajectory of the particle is given by 1

$$y = x \tan \theta - \frac{x^2 \sec^2 \theta}{2}.$$

- (ii) If there is only one path of trajectory for the particle to land at point P , find θ for that path. 4

Marks

- (c) Find the general solutions to the equation

6

$$\cos 4\theta + \cos 2\theta = \sqrt{2} \cos^2 \theta + \frac{1}{\sqrt{2}} \sin 2\theta.$$

Question 8 (15 marks)

Marked by SKB

Marks

- (a) (i) If $a > 0$, $b > 0$ and $c > 0$, show that $a^2 + b^2 \geq 2ab$ and hence deduce that $a^2 + b^2 + c^2 \geq ab + bc + ca$. 2
- (ii) If $a + b + c = 9$, show that $ab + bc + ca \leq 27$ and 3
- $$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \leq \frac{27}{abc}.$$
- (b) If $U_1 = 1, U_2 = 5$ and $U_n = 5U_{n-1} - 6U_{n-2}$ for $n \geq 3$, prove by mathematical induction that $U_n = 3^n - 2^n$ for $n \geq 1$. 5
- (c) The lines $y = 0$, $3x - 4y + 3 = 0$ and $3x + 4y - 15 = 0$ are the sides of a triangle. Find the co-ordinates of the centre of the circle inscribed in the triangle. Hence or otherwise write down the equation of the circle. 5

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

**MATHEMATICS EXTENSION 2
HIGHER SCHOOL CERTIFICATE
TRIAL EXAMINATION
SOLUTIONS
2008**

Question 1 (15 marks)

(a) $\int x \tan x^2 dx = \frac{1}{2} \int \tan u du$ where $u = x^2$
 (1 mark) $\frac{du}{dx} = 2x$
 $= \frac{1}{2} \int \frac{\sin u}{\cos u} du$
 $= -\frac{1}{2} \ln(\cos x^2) + c$
 (1 mark)

(b) $\int_4^9 \frac{x}{\sqrt{x}(1+x)} dx$ $u = \sqrt{x}, \frac{du}{dx} = \frac{1}{2\sqrt{x}} \therefore 2du = \frac{dx}{\sqrt{x}}$
 $= \int_2^3 \frac{u^2}{(1+u^2)} 2 \frac{du}{dx} dx$ $x=9, u=3$
 $= 2 \int_2^3 \frac{u^2}{1+u^2} du$ $x=4, u=2$
 (1 mark) for terminals
 $= 2 \int_2^3 \left(1 - \frac{1}{1+u^2}\right) du$ (1 mark) for function
 $= 2 \left[u - \tan^{-1} u \right]_2^3$
 $= 2 \left\{ (3 - \tan^{-1} 3) - (2 - \tan^{-1} 2) \right\}$
 $= 2 - 2 \tan^{-1} 3 + 2 \tan^{-1} 2$
 (1 mark)

$$\begin{aligned}
 \text{(c)} \quad & \int \frac{-2}{\sqrt{3+2x-x^2}} dx \\
 &= -2 \int \frac{1}{\sqrt{-(x^2-2x-3)}} dx \\
 &= -2 \int \frac{1}{\sqrt{-(x^2-2x+1-1-3)}} dx \\
 &= -2 \int \frac{1}{\sqrt{4-(x-1)^2}} dx \quad (1 \text{ mark}) \\
 &= -2 \sin^{-1} \frac{(x-1)}{2} + c \quad (1 \text{ mark})
 \end{aligned}$$

$u = x-1$
 $\frac{du}{dx} = 1$

$$\begin{aligned}
 \text{(d)} \quad \text{(i)} \quad & \frac{2x^2+2x+5}{(x^2+2)(1-x)} = \frac{ax+b}{x^2+2} + \frac{c}{1-x} \\
 &= \frac{(ax+b)(1-x)+c(x^2+2)}{(x^2+2)(1-x)} \\
 \text{True iff} \quad & 2x^2+2x+5 = (ax+b)(1-x)+c(x^2+2) \quad (1 \text{ mark}) \\
 & \text{Put } x=1, \quad 9=3c \quad c=3 \\
 & \text{Put } x=0, \quad 5=b+2c \quad b=-1 \\
 & \text{Put } x=-1, \quad 5=(-a-1)2+3 \times 3 \quad a=1 \\
 & \text{So, } a=1, \quad b=-1, \quad c=3 \\
 & \quad (1 \text{ mark})
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii)} \quad & \text{Hence } \int \frac{2x^2+2x+5}{(x^2+2)(1-x)} dx \\
 &= \int \left(\frac{x-1}{x^2+2} + \frac{3}{1-x} \right) dx \\
 &= \int \frac{x}{x^2+2} dx - \int \frac{1}{x^2+2} dx + \int \frac{3}{1-x} dx \quad (1 \text{ mark}) \\
 &= \frac{1}{2} \ln|x^2+2| - \frac{1}{\sqrt{2}} \tan^{-1} \frac{x}{\sqrt{2}} - 3 \ln|1-x| + c \\
 & \quad (or = \ln \left| \frac{\sqrt{x^2+2}}{(1-x)^3} \right| - \frac{1}{\sqrt{2}} \tan^{-1} \frac{x}{\sqrt{2}} + c) \quad (1 \text{ mark})
 \end{aligned}$$

$$(e) \quad \int_1^5 \frac{\ln x}{\sqrt{x}} dx = \left[2\sqrt{x} \ln x \right]_1^5 - \int_1^5 2\sqrt{x} \cdot \frac{1}{x} dx \quad \begin{array}{l} \text{Let } u = \ln x \quad dv = \frac{1}{\sqrt{x}} dx \\ \therefore \frac{du}{dx} = \frac{1}{x} \quad v = 2\sqrt{x} \end{array}$$

(1 mark) – first function (1 mark) – second function

(1 mark) – correct positioning of terminals

$$\begin{aligned} &= \left(2\sqrt{5} \ln 5 - 2 \ln 1 \right) - \int_1^5 2x^{-\frac{1}{2}} dx \\ &= 2\sqrt{5} \ln 5 - 2 \left[2\sqrt{x} \right]_1^5 \\ &= 2\sqrt{5} \ln 5 - 4\sqrt{5} + 4 \end{aligned}$$

(1 mark)

Question 2 (15 marks)

$$(a) \quad I = \int_{-1}^1 \frac{\tan^{-1} x}{1+x^4} dx \quad \text{Let } f(x) = \frac{\tan^{-1} x}{1+x^4}$$

$$\therefore f(-x) = \frac{\tan^{-1}(-x)}{1+(-x)^4} = -\frac{\tan^{-1} x}{1+x^4} = -f(x)$$

$$\therefore f(x) \text{ is an odd function} \quad (1 \text{ mark})$$

$\therefore I = 0$, as the integration of an odd function about symmetrical limits is zero. (1 mark)

$$(b) \quad I = \int_0^1 \sqrt{4-x^2} dx \quad \text{Let } x = 2 \sin \theta \therefore \frac{dx}{d\theta} = 2 \cos \theta$$

When $x = 0, \theta = 0$ and when $x = 1, \theta = \frac{\pi}{6}$

$$\therefore I = \int_0^{\frac{\pi}{6}} \sqrt{4-4\sin^2 \theta} \cdot 2 \cos \theta d\theta \quad (1 \text{ mark})$$

$$= 4 \int_0^{\frac{\pi}{6}} \cos^2 \theta d\theta \quad (1 \text{ mark})$$

$$= 2 \int_0^{\frac{\pi}{6}} 1 + \cos 2\theta d\theta, \text{ using } \cos 2\theta = 2\cos^2 \theta - 1 \text{ and } \cos^2 \theta = \frac{1}{2}[1 + \cos 2\theta]$$

$$= 2 \left[\theta + \frac{\sin 2\theta}{2} \right]_0^{\frac{\pi}{6}} \quad (1 \text{ mark})$$

$$= 2 \left[\frac{\pi}{6} + \frac{\sqrt{3}}{4} \right] \text{ or } \frac{2\pi + 3\sqrt{3}}{6} \text{ or } \frac{\pi}{3} + \frac{\sqrt{3}}{2} \quad (1 \text{ mark})$$

$$\begin{aligned}
 \text{(c)} \quad I &= \int_0^{2\pi} \frac{x \cos x}{1 + \sin^2 x} dx \\
 &= \int_0^{2\pi} \frac{(2\pi - x) \cos(2\pi - x)}{1 + \sin^2(2\pi - x)} dx, \text{ using the given result (1 mark)} \\
 &= \int_0^{2\pi} \frac{2\pi \cos x}{1 + \sin^2 x} dx - \int_0^{2\pi} \frac{x \cos x}{1 + \sin^2 x} dx \\
 \therefore 2I &= \int_0^{2\pi} \frac{2\pi \cos x}{1 + \sin^2 x} dx \\
 \therefore I &= \pi \int_0^{2\pi} \frac{\cos x}{1 + \sin^2 x} dx \quad \text{(1 mark)} \quad \text{Let } u = \sin x, \therefore \frac{du}{dx} = \cos x \\
 &\quad \text{When } x = 0, u = 0 \text{ and when } x = 2\pi, u = 0 \\
 \therefore I &= \pi \int_0^0 \frac{du}{1 + u^2} = 0, \text{ as the integration about the same limits is zero. (1 mark)}
 \end{aligned}$$

$$\begin{aligned}
 \text{(d)} \quad \text{(i)} \quad I_n &= \int_0^1 x(x^2 - 1)^n dx \quad n = 0, 1, 2, \dots \\
 &= \left[\frac{x^2}{2} (x^2 - 1)^n \right]_0^1 - \int_0^1 \frac{x^2}{2} \times n(x^2 - 1)^{n-1} \times 2x dx \quad \text{(1 mark)} \\
 &= \frac{1}{2} \times 0 - 0 - n \int_0^1 x^3 (x^2 - 1)^{n-1} dx \\
 &= -n \int_0^1 \frac{x^3 (x^2 - 1)^n}{x^2 - 1} dx \\
 &= -n \int_0^1 \frac{[x(x^2 - 1) + x](x^2 - 1)^n}{x^2 - 1} dx \\
 &= -n \int_0^1 \left\{ x(x^2 - 1)^n + \frac{x}{x^2 - 1} (x^2 - 1)^n \right\} dx \quad \text{(1 mark)} \\
 I_n &= -n \int_0^1 x(x^2 - 1)^n dx - n \int_0^1 x(x^2 - 1)^{n-1} dx \\
 I_n &= -nI_n - nI_{n-1} \\
 (1+n)I_n &= -nI_{n-1} \\
 \text{So } I_n &= \frac{-n}{n+1} I_{n-1} \quad \text{for } n \geq 1 \quad \text{as required. (1 mark)}
 \end{aligned}$$

(ii) Method 1 – “Hence”

$$I_n = \frac{-n}{n+1} I_{n-1} \quad \text{for } n \geq 1$$

$$= \frac{-n}{n+1} \frac{-n+1}{n} \frac{-n+2}{n-1} \dots \frac{-3}{4} \frac{-2}{3} \frac{-1}{2} I_0$$

$$\text{Now } I_0 = \int_0^1 x(x^2-1)^0 dx$$

$$= \left[\frac{x^2}{2} \right]_0^1$$

$$= \frac{1}{2} \quad (1 \text{ mark})$$

$$\text{So } I_n = \frac{-n}{n+1} \frac{-n+1}{n} \frac{-n+2}{n-1} \dots \frac{-3}{4} \frac{-2}{3} \frac{-1}{2} \frac{1}{2} \text{ for } n \geq 0$$

$$= (-1)^n \frac{n}{n+1} \frac{n-1}{n} \frac{n-2}{n-1} \dots \frac{3}{4} \frac{2}{3} \frac{1}{2} \frac{1}{2}$$

$$= (-1)^n \frac{1}{n+1} \cdot \frac{1}{2} \quad (\text{The other terms cancel.}) \quad (1 \text{ mark})$$

$$\text{So } I_n = \frac{(-1)^n}{2(n+1)}, \quad n \geq 0 \quad \text{as required.}$$

OR

$$I_n = \frac{-n}{n+1} I_{n-1} \quad \text{for } n \geq 1$$

$$I_n = \frac{-n}{n+1} \int_0^1 x(x^2-1)^{n-1} dx$$

$$= \frac{-n}{n+1} \frac{1}{2} \int_0^1 2x(x^2-1)^{n-1} dx$$

$$= \frac{-n}{n+1} \frac{1}{2} \left[\frac{(x^2-1)^n}{n} \right]_0^1 \quad (2 \text{ marks})$$

$$= \frac{-n}{n+1} \frac{1}{2} \left[0 - \frac{(-1)^n}{n} \right]$$

$$\text{So } I_n = \frac{(-1)^n}{2(n+1)}, \quad n \geq 0 \quad \text{as required.}$$

(ii) Method 2 – “Otherwise”

$$\begin{aligned}
 I_n &= \int_0^1 x(x^2 - 1)^n dx \text{ for } n = 0, 1, 2, \dots \\
 &= \frac{1}{2} \int_0^1 2x(x^2 - 1)^n dx \\
 &= \frac{1}{2} \left[\frac{(x^2 - 1)^{n+1}}{n+1} \right]_0^1 \quad (1 \text{ mark}) \\
 &= \frac{1}{2(n+1)} (0 - (-1)^{n+1}) \\
 &= \frac{-1(-1)^{n+1}}{2(n+1)} \\
 &= \frac{(-1)^n}{2(n+1)} \text{ for } n \geq 0 \text{ as required} \quad (1 \text{ mark})
 \end{aligned}$$

$$(iii) \quad I_0 = \frac{1}{2}, I_1 = \frac{-1}{4}, I_2 = \frac{1}{6}, I_3 = \frac{-1}{8}, I_4 = \frac{1}{10}, I_5 = \frac{-1}{12}$$

Clearly $I_{2n} > 0$ and $I_{2n+1} < 0$ So $I_{2n} > I_{2n+1}$

$$\begin{aligned}
 \text{Alternatively, from (ii), } I_{2n} &= \frac{(-1)^{2n}}{2(2n+1)} \\
 &= \frac{1}{2(2n+1)} \\
 &> 0 \\
 I_{2n+1} &= \frac{(-1)^{2n+1}}{2((2n+1)+1)} \\
 &= \frac{(-1)}{4(n+1)} \\
 &< 0
 \end{aligned}$$

So $I_{2n} > I_{2n+1}$.

(1 mark)

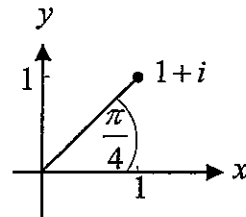
Question 3 (15 marks)

$$\begin{aligned}
 (a) \quad (i) \quad z\bar{w} &= (3-i)(2-4i) \\
 &= 6 - 12i - 2i - 4 \\
 &= 2 - 14i \quad (1 \text{ mark})
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii)} \quad \frac{z}{w} &= \frac{3-i}{2+4i} \times \frac{2-4i}{2-4i} \\
 &= \frac{2-14i}{20} \\
 &= \frac{1}{10} - \frac{7}{10}i
 \end{aligned}$$

(1 mark)

(b) (i) From the diagram,
 $1+i = \sqrt{2} \operatorname{cis} \frac{\pi}{4}$



(1 mark) for correct modulus
(1 mark) for correct argument

(ii) $(1+i)^n + (1-i)^n = 0$

Hence

$$\left(\sqrt{2}cis\frac{\pi}{4}\right)^n + \left(\sqrt{2}cis\frac{-\pi}{4}\right)^n = 0 \quad (1 \text{ mark})$$

$$(\sqrt{2})^n \left(\cos\frac{\pi n}{4} + i\sin\frac{\pi n}{4}\right) + (\sqrt{2})^n \left(\cos\frac{\pi n}{4} - i\sin\frac{\pi n}{4}\right) = 0$$

$$\cos\frac{\pi n}{4} = 0 \quad (1 \text{ mark})$$

$$\frac{\pi n}{4} = \frac{(2k+1)\pi}{2}$$

where $k = 0, 1, 2, \dots$

$$n = 2(2k+1)$$

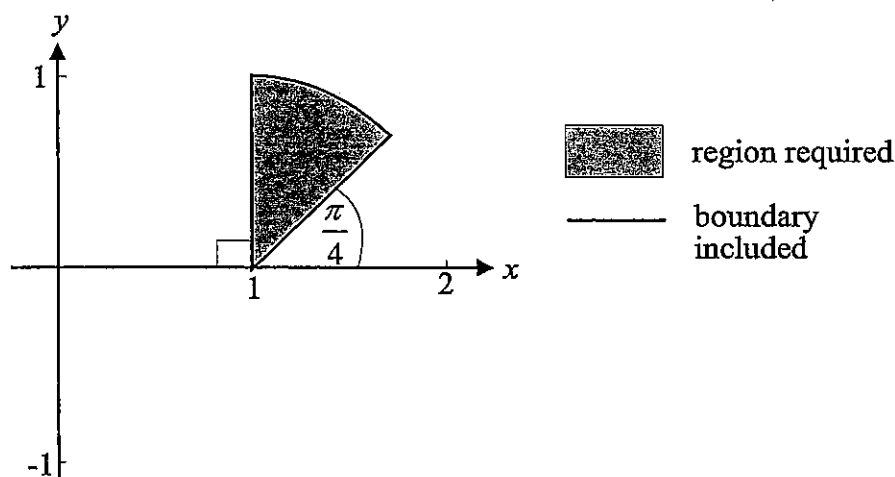
(1 mark)

- (c) The inequality $|z-1| \leq 1$ corresponds to a disc with centre at $(1,0)$ and radius 1.

The inequality $\frac{\pi}{4} \leq \arg(z-1) \leq \frac{\pi}{2}$ corresponds to a wedge with vertex $(1,0)$.

(1 mark)

The region where both these inequalities hold is shown in the diagram below.



(1 mark)

- (d) Point P corresponds to the complex number z . Point Q is obtained by rotating point P **clockwise** through an angle of $\frac{\pi}{2}$ and reducing it by a factor of $\frac{1}{3}$.

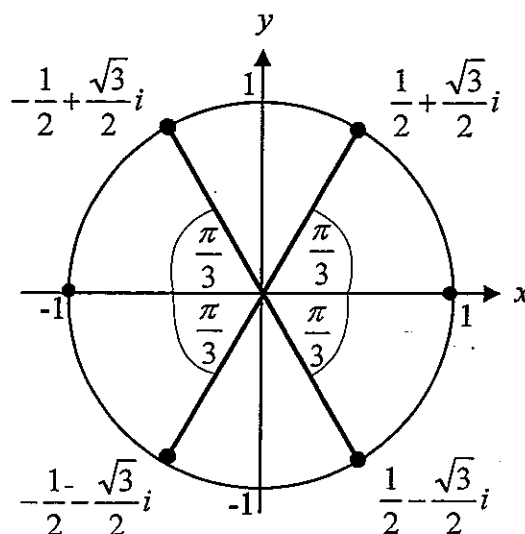
So, point Q corresponds to the complex number $\frac{-i}{3}z$.

(1 mark)

(e) (i) $z^6 = 1$

We are looking for the sixth roots of unity. We know that one root is 1 and another is -1 . The 6 roots of unity are evenly spaced around the circumference of a circle of radius 1 unit.

So, the other four must be $\text{cis}\frac{\pi}{3}$, $\text{cis}\frac{2\pi}{3}$, $\text{cis}\frac{4\pi}{3}$ and $\text{cis}\frac{5\pi}{3}$.



(1 mark)

So, the six roots are

$$\pm 1, +\frac{1}{2} \pm \frac{\sqrt{3}}{2}i, -\frac{1}{2} \pm \frac{\sqrt{3}}{2}i. \quad (1 \text{ mark})$$

(ii) $z^6 - 1 = 0$

$$(z^3 - 1)(z^3 + 1) = 0$$

$$(z - 1)(z^2 + z + 1)(z + 1)(z^2 - z + 1) = 0 \quad (1 \text{ mark})$$

The two real roots of the equation are revealed by the factors $(z - 1)$ and $(z + 1)$. The four non-real roots are revealed by the factors $(z^2 + z + 1)$ and $(z^2 - z + 1)$.

$$\text{So } (\omega^2 + \omega + 1)(\omega^2 - \omega + 1) = 0$$

$$\omega^4 + \omega^2 + 1 = 0$$

$$\text{So } \omega^4 + \omega^2 = -1 \quad \text{as required.} \quad (1 \text{ mark})$$

(iii) Let $\omega = \text{cis}\left(\frac{\pi}{3}\right)$

Now, $\omega^4 = \text{cis}\frac{4\pi}{3}$ (De Moivre)

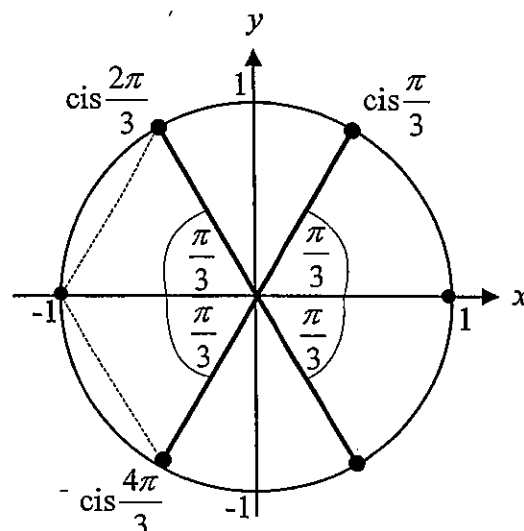
$\omega^2 = \text{cis}\frac{2\pi}{3}$ (De Moivre)

$\text{cis}\frac{4\pi}{3} + \text{cis}\frac{2\pi}{3} = -1$ by adding

the two complex numbers $\text{cis}\frac{4\pi}{3}$ and $\text{cis}\frac{2\pi}{3}$.

(Note, any of the four possible values of ω could have been chosen here to illustrate that $\omega^4 + \omega^2 = -1$.)

(1 mark)



Question 4 (15 marks)

(a) (i) P is the point $(a \cos \theta, b \sin \theta)$

$x = a \cos \theta$ $y = b \sin \theta$

$\frac{dx}{d\theta} = -a \sin \theta$ $\frac{dy}{d\theta} = b \cos \theta$

$\frac{dy}{dx} = -\frac{b \cos \theta}{a \sin \theta}$

(1 mark)

Equation of tangent is

$y - b \sin \theta = -\frac{b \cos \theta}{a \sin \theta} (x - a \cos \theta)$

$ay \sin \theta - ab \sin^2 \theta = -bx \cos \theta + ab \cos^2 \theta$

$bx \cos \theta + ay \sin \theta = ab(\sin^2 \theta + \cos^2 \theta)$

$\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1$

(1 mark)

as required.

- (ii) If T is the point of intersection between the tangent found in part (i) and one of the directrices of the ellipse, then T has the coordinates $\left(\frac{a}{e}, 0\right)$. **(1 mark)**

From (i), the gradient of the tangent at P is $\frac{-b \cos \theta}{a \sin \theta}$. So using the coordinates of points P and T we have

$$\frac{b \sin \theta - 0}{a \cos \theta - \frac{a}{e}} = \frac{-b \cos \theta}{a \sin \theta} \quad \text{(1 mark)}$$

$$\frac{eb \sin \theta}{ae \cos \theta - a} = \frac{-b \cos \theta}{a \sin \theta}$$

$$be \sin \theta \times a \sin \theta = -b \cos \theta (ae \cos \theta - a)$$

$$abe \sin^2 \theta = -abe \cos^2 \theta + ab \cos \theta$$

$$abe(\sin^2 \theta + \cos^2 \theta) = ab \cos \theta$$

$$\cos \theta = e \quad \text{as required.}$$

(1 mark)

- (iii) Since $\cos \theta = e$, the x -coordinate of P which is $a \cos \theta = ae$. So the focal chord through P makes an angle of 90° with the x -axis. **(1 mark)**

- (iv) The equation of the tangent through P is

$$\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1$$

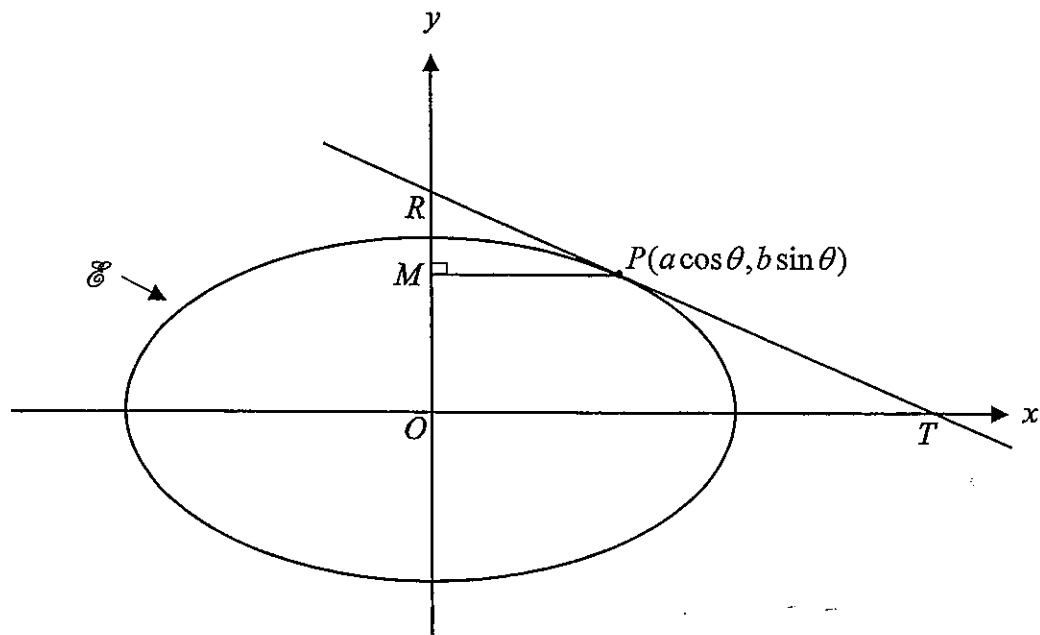
when $x = 0$,

$$y \sin \theta = b$$

$$y = \frac{b}{\sin \theta}$$

R is the point $\left(0, \frac{b}{\sin \theta}\right)$

(1 mark)



Let M be the point on the y -axis such that PM is perpendicular to the y -axis.

M is the point $(0, b \sin \theta)$

Now since $\triangle ROT$ is similar to $\triangle RMP$,

$$\frac{RP}{RT} = \frac{RM}{RO} \quad (1 \text{ mark})$$

$$= \frac{\frac{b}{\sin \theta} - b \sin \theta}{\frac{b}{\sin \theta}}$$

$$= \frac{b - b \sin^2 \theta}{\sin \theta} \times \frac{\sin \theta}{b}$$

$$= 1 - \sin^2 \theta$$

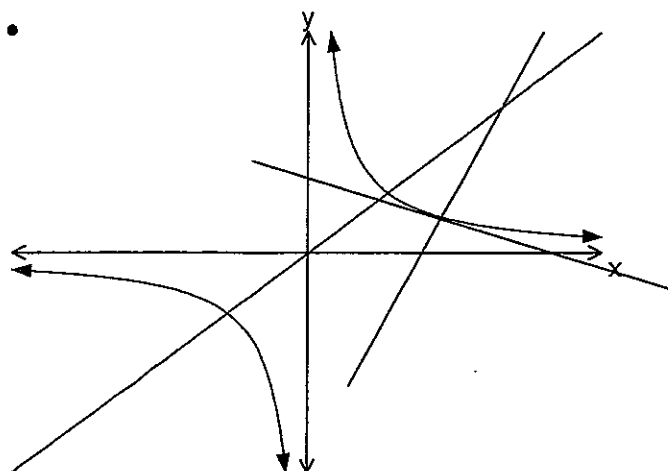
$$= \cos^2 \theta$$

$$= e^2 \quad (\text{from part (ii)})$$

So $RP = e^2 RT$ as required.

(1 mark)

(b) (1 mark)



: x Intercept (0, 0)

$$xy = c^2 \therefore y = c^2 x^{-1} \text{ and } \frac{dy}{dx} = -\frac{c^2}{x^2}$$

$$\text{At } P(ct, \frac{c}{t}) \quad \frac{dy}{dx} = \frac{-c^2}{c^2 t^2} = \frac{-1}{t^2} = m_{\text{tangent}}; m_{\text{normal}} = t^2 \quad (1 \text{ mark})$$

Now the equation of the tangent at P is:

$$y - \frac{c}{t} = \frac{-1}{t^2}(x - ct)$$

$$\therefore t^2 y - ct = -x + ct$$

$$\therefore x + t^2 y = 2ct$$

$$\text{At T } y = x \therefore x + t^2 x = 2ct \therefore x = \frac{2ct}{1+t^2}$$

$$\Rightarrow T = (\frac{2ct}{1+t^2}, \frac{2ct}{1+t^2}) \quad (1 \text{ mark})$$

Now the equation of the normal at P is:

$$y - \frac{c}{t} = t^2(x - ct)$$

$$\therefore t^3 x - ty = c(t^4 - 1)$$

$$\text{At N } y = x \therefore t^3 x - tx = c(t^4 - 1) \therefore x = \frac{c(t^4 - 1)}{t(t^2 - 1)} = \frac{c(t^2 + 1)}{t}$$

$$\Rightarrow N = (\frac{c(t^2 + 1)}{t}, \frac{c(t^2 + 1)}{t}) \quad (1 \text{ mark})$$

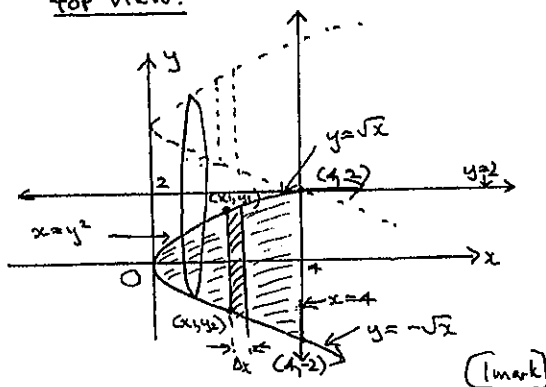
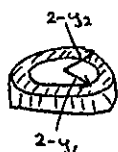
$$\text{Now } OT \cdot ON = \sqrt{(\frac{2ct}{1+t^2})^2 + (\frac{2ct}{1+t^2})^2} \cdot \sqrt{(\frac{c(t^2 + 1)}{t})^2 + (\frac{c(t^2 + 1)}{t})^2}$$

$$= \sqrt{2}(\frac{2ct}{1+t^2}) \cdot \sqrt{2}(\frac{c(t^2 + 1)}{t})$$

$$= 4c^2$$

(2 marks)

5 (a)

TOP VIEW:Take slice of thickness $\Delta x \perp$ to x -axisSIDE VIEW:

$$y_1 = \sqrt{x}, R_2 = 2 - y_2$$

$$y_2 = -\sqrt{x}, R_1 = 2 - y_1$$

Area of cross-sectional slice, $A(x) = \pi(R_2^2 - R_1^2)$

$$\therefore A(x) = \pi((2 - \sqrt{x})^2 - (2 + \sqrt{x})^2)$$

$$= \pi(4 - 4\sqrt{x} + x - (4 + 4\sqrt{x} + x))$$

$$= 8\pi\sqrt{x} \quad (1 \text{ mark})$$

$$\text{Volume, } \Delta V, \text{ of each slice} = A(x)\Delta x$$

$$\text{Now total volume} = \lim_{\Delta x \rightarrow 0} \sum_{x=0}^4 A(x)\Delta x$$

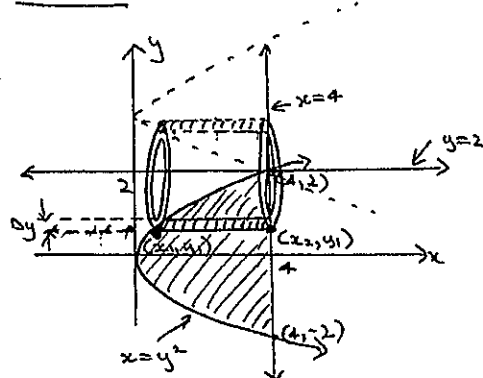
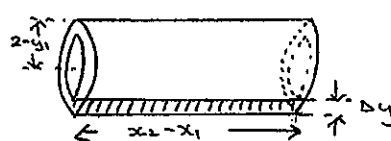
$$= 8\pi \int_0^4 x^{\frac{1}{2}} dx \quad (1 \text{ mark})$$

$$= 8\pi \left[\frac{x^{\frac{3}{2}}}{\frac{3}{2}} \right]_0^4$$

$$= \frac{16\pi}{3} [4^{\frac{3}{2}} - 0]$$

$$= \frac{128\pi}{3} \text{ units}^3 \quad (1 \text{ mark})$$

(b)

TOP VIEW:Take slice of thickness $\Delta y \perp$ to y -axisSIDE VIEW:Each slice when rotated about the line $y = 2$ generates a thin cylindrical shell of area $A(y) = 2\pi rh$.Volume, ΔV , of each slice $= A(y)\Delta y$

$$\text{Now total volume} = \lim_{\Delta y \rightarrow 0} \sum_{y=-2}^2 A(y)\Delta y$$

$$\text{where } A(y) = 2\pi(2 - y)(4 - y^2) \quad (1 \text{ mark})$$

$$\therefore \text{Total volume} = 2\pi \int_{-2}^2 (8 - 4y - 2y^2 + y^3) dy$$

$$(1 \text{ mark})$$

$$= 2\pi \left[8y - 2y^2 - \frac{2y^3}{3} + \frac{y^4}{4} \right]_{-2}^2$$

$$= 2\pi \left[(16 - 8 - \frac{16}{3} + 4) - (-16 - 8 + \frac{16}{3} + 4) \right]$$

$$= 2\pi \left[32 - \frac{16}{3} \times 2 \right]$$

$$(1 \text{ mark}) \quad = \frac{128\pi}{3} \text{ units}^3$$

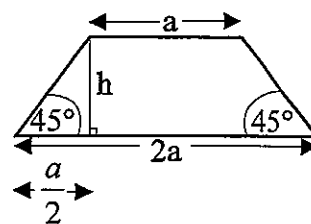
(c) (i) $\tan 45^\circ = h \div \frac{a}{2}$

$$h = \frac{a}{2}$$

Area of trapezium at $x = b$ is given by

$$\begin{aligned} & \frac{1}{2}(a+2a)\frac{a}{2} \\ &= \frac{3a^2}{4} \end{aligned}$$

(1 mark)



(ii) Consider a slice of width Δx

$$\Delta V = \frac{3a^2}{4} \Delta x \quad \text{where } 0 \leq x \leq 1$$

$$= \frac{3}{4} y^2 \Delta x$$

$$= \frac{3}{4} (1-x^2) \Delta x \quad \text{since } x^2 + y^2 = 1 \quad (1 \text{ mark})$$

$$\text{So Volume} = \lim_{\Delta x \rightarrow 0} \sum_{x=0}^1 \frac{3}{4} (1-x^2) \Delta x$$

$$= \frac{3}{4} \int_0^1 (1-x^2) dx$$

$$= \frac{3}{4} \left[x - \frac{x^3}{3} \right]_0^1$$

$$= \frac{3}{4} \left(\frac{2}{3} \right)$$

$$= \frac{1}{2} \text{ unit}^3$$

(1 mark)

(iii) The solid generated when the semicircular base of $\mathcal{D}$ is rotated through an angle of 90° is a quarter of a sphere. Hence the volume is

$$\frac{4}{3} \pi r^3 \div 4. \text{ So required volume is } \frac{\pi}{3} \text{ since } r = 1.$$

(1 mark)

(iv) From part (i), area of a trapezium at $x = b$ is given by $\frac{1}{2}(a+2a)h$ where h is the height of the trapezium.

$$\begin{aligned}\text{Now } \tan \theta &= h \div \frac{a}{2} \\ &= \frac{2h}{a}\end{aligned}$$

So area of trapezium at $x = b$

$$\begin{aligned}&= \frac{3a}{2} \times \frac{a}{2} \tan \theta \\ &= \frac{3a^2}{4} \tan \theta\end{aligned}$$

$$\begin{aligned}\text{So Volume} &= \frac{3}{4} \tan \theta \int_0^1 (1-x^2) dx \quad \text{where } \theta \text{ is constant for a particular solid} \\ &= \frac{1}{2} \tan \theta \quad \text{(1 mark)}\end{aligned}$$

We want to find θ such that

$$\begin{aligned}\frac{1}{2} \tan \theta &> \frac{\pi}{3} \\ \tan \theta &> \frac{2\pi}{3} \\ \theta &> 64^\circ 29' \quad \text{(1 mark)}\end{aligned}$$

However $\theta < 90^\circ$ since we have a trapezium.

So we require $64^\circ 29' < \theta < 90^\circ$. (1 mark)

Question 6 (15 marks)

- (a) (i) As the coefficients of $P(x)$ are real then $1+i$ is a further root of $P(x)$.

$$\begin{aligned}\therefore P(x) &= (x-1+i)(x-1-i)R(x) \\ &= ([x-1]^2 - i^2)R(x) \quad \text{(1mark)} \\ &= (x^2 - 2x + 2)R(x)\end{aligned}$$

As $1-i$ is a root then $P(1-i) = 0$

$$\begin{aligned}\therefore (1-i)^3 + a(1-i)^2 + b(1-i) + 6 &= 0 \\ \therefore 1 + 3(-i) + 3(-i)^2 + (-i)^3 + a(1-2i+i^2) + b-bi + 6 &= 0 \quad \text{(1mark)} \\ \therefore 1-3i-3+i+a-2ai-a+b-bi+6 &= 0 \\ \therefore 4+b+i(-2-2a-b) &= 0\end{aligned}$$

Equating real and imaginary parts:

$$4+b=0 \dots\dots\dots(1) \quad \text{(1mark)}$$

$$-2-2a-b=0 \dots\dots\dots(2)$$

$$\text{From (1) } b = -4 \text{ sub. into (2) } \therefore a = 1 \quad \text{(1mark)}$$

$$\begin{aligned}\text{(ii) } \therefore P(x) &= x^3 + x^2 - 4x + 6 \\ &= (x^2 - 2x + 2)(x+3) \\ &= (x-1+i)(x-1-i)(x+3) \text{ over the complex field.} \\ &\quad \text{(1mark)}\end{aligned}$$

(b) $(\cos \theta + i \sin \theta)^5 = \cos 5\theta + i \sin 5\theta$ (De Moivre)

$$\begin{aligned}
 (\cos \theta + i \sin \theta)^5 &= \cos^5 \theta + 5 \cos^4 \theta (i \sin \theta) + 10 \cos^3 \theta (i \sin \theta)^2 + 10 \cos^2 \theta (i \sin \theta)^3 \\
 &\quad + 5 \cos \theta (i \sin \theta)^4 + (i \sin \theta)^5 \\
 &= \cos^5 \theta + i 5 \cos^4 \theta \sin \theta - 10 \cos^3 \theta \sin^2 \theta - i 10 \cos^2 \theta \sin^3 \theta \\
 &\quad + 5 \cos \theta \sin^4 \theta + i \sin^5 \theta
 \end{aligned}$$

(1 mark)

By equating real and imaginary parts,

$$\cos 5\theta = \cos^5 \theta - 10 \cos^3 \theta \sin^2 \theta + 5 \cos \theta \sin^4 \theta$$

$$\sin 5\theta = 5 \cos^4 \theta \sin \theta - 10 \cos^2 \theta \sin^3 \theta + \sin^5 \theta$$

(1 mark)

$$\begin{aligned}
 \tan 5\theta &= \frac{\sin 5\theta}{\cos 5\theta} \\
 &= \frac{5 \cos^4 \theta \sin \theta - 10 \cos^2 \theta \sin^3 \theta + \sin^5 \theta}{\cos^5 \theta - 10 \cos^3 \theta \sin^2 \theta + 5 \cos \theta \sin^4 \theta} \\
 &= \frac{\frac{5 \sin \theta}{\cos \theta} - \frac{10 \sin^3 \theta}{\cos^3 \theta} + \frac{\sin^5 \theta}{\cos^5 \theta}}{1 - \frac{10 \sin^2 \theta}{\cos^2 \theta} + \frac{5 \sin^4 \theta}{\cos^4 \theta}}
 \end{aligned}$$

$$\text{So } \tan 5\theta = \frac{5 \tan \theta - 10 \tan^3 \theta + \tan^5 \theta}{1 - 10 \tan^2 \theta + 5 \tan^4 \theta}$$

(1 mark)

(ii)

Let $x = \tan \theta$ and $\tan 5\theta = 1$

$$\text{So, } \tan 5\theta = \frac{5 \tan \theta - 10 \tan^3 \theta + \tan^5 \theta}{1 - 10 \tan^2 \theta + 5 \tan^4 \theta}$$

becomes $\frac{5x - 10x^3 + x^5}{1 - 10x^2 + 5x^4} = 1$

$$\text{So, } x^5 - 5x^4 - 10x^3 + 10x^2 + 5x - 1 = 0$$

For $\tan 5\theta = 1$

$$5\theta = n\pi + \frac{\pi}{4}, \quad n \text{ is an integer}$$

$$\theta = \frac{n\pi}{5} + \frac{\pi}{20}$$

$$n = 0, \theta = \frac{\pi}{20}$$

$$n = 1, \theta = \frac{\pi}{4}$$

$$n = -1, \theta = \frac{-3\pi}{20}$$

$$n = 2, \theta = \frac{9\pi}{20}$$

$$n = -2, \theta = \frac{-7\pi}{20}$$

$$\text{So, } x = \tan \frac{\pi}{4} = 1, \tan \frac{\pi}{20}, \tan \frac{9\pi}{20}, \tan \left(\frac{-3\pi}{20} \right) = -\tan \frac{3\pi}{20} \text{ and } \tan \left(\frac{-7\pi}{20} \right) = -\tan \frac{7\pi}{20}$$

(2 marks)

(iii)

$$\text{let } P(x) = x^5 - 5x^4 - 10x^3 + 10x^2 + 5x - 1 = 0$$

$$(x-1)(x^4 - 4x^3 - 14x^2 - 4x + 1) = 0 \quad (1 \text{ mark})$$

$$x^4 - 4x^3 - 14x^2 - 4x + 1 = x^2 \left(x^2 - 4x - 14 - \frac{4}{x} + \frac{1}{x^2} \right) = 0$$

$$\text{So, } x^2 \left(x^2 + \frac{1}{x^2} - 4 \left(x + \frac{1}{x} \right) - 14 \right) = 0$$

$$\text{but } x^2 \neq 0 \text{ so, } \left(x + \frac{1}{x} \right)^2 - 4 \left(x + \frac{1}{x} \right) - 16 = 0$$

$$\begin{aligned} \text{since } P(0) \neq 0, \quad x + \frac{1}{x} &= \frac{4 \pm \sqrt{16 - 4(1)(-16)}}{2} \\ &= \frac{4 \pm \sqrt{80}}{2} \\ &= \frac{4 \pm 4\sqrt{5}}{2} \end{aligned}$$

$$\text{So, } x + \frac{1}{x} = 2 + 2\sqrt{5}, \text{ or } x + \frac{1}{x} = 2 - 2\sqrt{5}$$

(2 marks)

Now $\tan \theta = \cot \left(\frac{\pi}{2} - \theta \right) = \frac{1}{\tan \left(\frac{\pi}{2} - \theta \right)},$

so, if $x = \tan \frac{\pi}{20}$ then $\tan \frac{\pi}{20} + \frac{1}{\tan \frac{\pi}{20}} = 2 + 2\sqrt{5}$

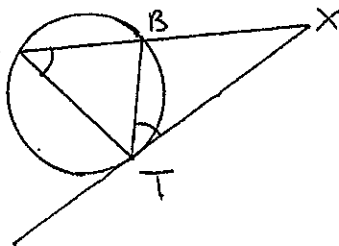
and hence $\tan \frac{\pi}{20} + \tan \frac{9\pi}{20} = 2 + 2\sqrt{5}$

Similarly, if $x = -\tan \frac{3\pi}{20}$ then $-\tan \frac{3\pi}{20} - \frac{1}{\tan \frac{3\pi}{20}} = 2 - 2\sqrt{5}$

and hence $\tan \frac{3\pi}{20} + \tan \frac{7\pi}{20} = 2\sqrt{5} - 2$ **(2 marks)**

Question 7

(a) (i) A

In Δ s XAT and XTB:

$$\angle XAT = \angle XTB$$

[Angle between tangent and chord at point of contact equals angle in alternate segment] [1 mark]

$\angle X$ is common

$$\angle XTA = \angle XBT \quad \left[\begin{array}{l} \text{Remaining } \angle\text{s} \\ \text{are equal;} \\ \angle \text{sum of } \Delta = 180^\circ \end{array} \right]$$

$\therefore$ As Δ s are equiangular

$$\therefore \Delta XAT \sim \Delta XTB \quad [1 \text{ mark}]$$

(ii) Now $\frac{XA}{XT} = \frac{XT}{XB}$ [Corresponding sides of similar Δ s are in the same ratio.] [1 mark]

$$\therefore XT^2 = XA \cdot XB. \quad [1 \text{ mark}]$$

(b)

$$(i) \quad x = \sqrt{g} t \cos \theta$$

$$\text{So} \quad t = \frac{x}{\sqrt{g} \cos \theta}$$

$$\text{In} \quad y = \sqrt{g} t \sin \theta - \frac{1}{2} g t^2$$

$$\text{becomes } y = \frac{\sqrt{g} x \sin \theta}{\sqrt{g} \cos \theta} - \frac{1}{2} g \frac{x^2}{g \cos^2 \theta}$$

$$\text{So,} \quad y = x \tan \theta - \frac{x^2 \sec^2 \theta}{2}$$

(1 mark)

(ii) From (i) we have

$$y = x \tan \theta - \frac{x^2 \sec^2 \theta}{2}$$

$$= x \tan \theta - \frac{x^2}{2} (1 + \tan^2 \theta)$$

At point P, $x = d \cos 30^\circ$ and $y = d \sin 30^\circ$

$$= \frac{\sqrt{3}d}{2} \quad = \frac{d}{2}$$

(1 mark)

$$\text{So} \quad \frac{d}{2} = \frac{\sqrt{3}d}{2} \tan \theta - \frac{3d^2}{8} (1 + \tan^2 \theta)$$

$$4d = 4\sqrt{3}d \tan \theta - 3d^2 - 3d^2 \tan^2 \theta$$

$$3d^2 \tan^2 \theta - 4\sqrt{3}d \tan \theta + 3d^2 + 4d = 0$$

(1 mark)

We have a quadratic in $\tan \theta$.

$$\begin{aligned} \text{So} \quad \Delta &= 48d^2 - 4 \times 3d^2 (3d^2 + 4d) \\ &= 48d^2 - 12d^2 (3d^2 + 4d) \\ &= 12d^2 (4 - 3d^2 - 4d) \end{aligned}$$

If there is one path of trajectory for the particle to land at point P then $\Delta = 0$.

So $12d^2(4-3d^2-4d)=0$
 $4-3d^2-4d=0$ ($12d^2 \neq 0$)
 $(-3d+2)(d+2)=0$
 $d=\frac{2}{3}$ or $d=-2$ reject this since $d > 0$
 So $d=\frac{2}{3}$ (1 mark)
 So we have, $\frac{4}{3}\tan^2\theta - \frac{8\sqrt{3}}{3}\tan\theta + 4 = 0$
 So $\tan\theta = \left(\frac{8\sqrt{3}}{3} \pm \sqrt{0}\right) \div \frac{8}{3}$
 $= \sqrt{3}$
 $\theta = 60^\circ$ (1 mark)

(c)

$$\cos 4\theta + \cos 2\theta = \sqrt{2}\cos^2\theta + \frac{1}{\sqrt{2}}\sin 2\theta$$

$$2\cos 3\theta \cos \theta = \sqrt{2}\cos^2\theta + \sqrt{2}\sin\theta \cos\theta$$
(1 mark)

$$\sqrt{2}\cos 3\theta \cos \theta = \cos\theta(\cos\theta + \sin\theta)$$

$$= \cos\theta\left(\sqrt{2}\cos\left(\theta - \frac{\pi}{4}\right)\right)$$
(1 mark)

$$\text{So } \cos 3\theta \cos \theta = \cos\theta \cos\left(\theta - \frac{\pi}{4}\right)$$

$$\cos\theta = 0 \text{ or } \cos 3\theta = \cos\left(\theta - \frac{\pi}{4}\right)$$
(1 mark)

$$\theta = 2n\pi \pm \frac{\pi}{2} \text{ or } 3\theta = 2n\pi \pm \left(\theta - \frac{\pi}{4}\right) \quad n \text{ is an integer}$$

$$2\theta = 2n\pi - \frac{\pi}{4} \quad \text{or} \quad 4\theta = 2n\pi + \frac{\pi}{4}$$

So, $\theta = 2n\pi \pm \frac{\pi}{2}$ (1 mark) or $\theta = n\pi - \frac{\pi}{8}$ (1 mark) or $\theta = \frac{n\pi}{2} + \frac{\pi}{16}$ (1 mark)

Question 8 (15 marks)

(a) (i) $(a-b)^2 \geq 0$

$$a^2 + b^2 - 2ab \geq 0$$

$$a^2 + b^2 \geq 2ab \quad (\text{equality iff } a = b) \quad (1 \text{ mark})$$

Similarly, $b^2 + c^2 \geq 2bc$

$$a^2 + c^2 \geq 2ac$$

By addition, $2(a^2 + b^2 + c^2) \geq 2(ab + bc + ca)$

$$a^2 + b^2 + c^2 \geq ab + bc + ca \quad (\text{equality iff } a = b = c)$$

(1 mark)

(ii) $(a+b+c)^2 = a^2 + b^2 + c^2 + 2(ab + ac + bc)$

Since $a^2 + b^2 + c^2 \geq ab + bc + ca$

$$(a+b+c)^2 \geq ab + bc + ca + 2(ab + ac + bc)$$

$$(a+b+c)^2 \geq 3(ab + bc + ca)$$

Since $a + b + c = 9$,

$$81 \geq 3(ab + bc + ca)$$

$$ab + bc + ca \leq 27 \quad (\text{equality iff } a = b = c) \quad (2 \text{ marks})$$

$$\frac{1}{abc}(ab + bc + ca) \leq \frac{27}{abc}$$

$$\frac{1}{c} + \frac{1}{a} + \frac{1}{b} \leq \frac{27}{abc}$$

$$\text{ie } \frac{1}{a} + \frac{1}{b} + \frac{1}{c} \leq \frac{27}{abc} \quad \text{as required} \quad (1 \text{ mark})$$

8(c)

[illegible]

Let $A = (-1, 0)$, $C = (5, 0)$

$$\begin{aligned} 3x - 4y + 3 &= 0 & \text{--- (1)} \\ 3x + 4y - 15 &= 0 & \text{--- (2)} \end{aligned}$$
$$\textcircled{1} + \textcircled{2}: \quad 6x - 12 = 0 \quad \therefore x = 2 \text{ substituting } \textcircled{2}$$

$$\therefore y = 2\frac{1}{2} \quad \therefore B = (2, 2\frac{1}{2})$$
$$\text{Now } d_{AB} = \sqrt{(2-1)^2 + (2\frac{1}{4})^2} = 3\frac{3}{4}$$
$$d_{BC} = \sqrt{(5-2)^2 + (-2\frac{1}{4})^2} = 3\frac{3}{4}$$

$\therefore x=2$ is the right bisector [mark]
of side AC.

Let X, Y and Z be the feet of the perpendiculars drawn from P to each line.

$$\therefore |y_1| = \frac{|6+4y_1-15|}{\sqrt{3^2+4^2}} = \frac{|6-4y_1+3|}{\sqrt{3^2+4^2}} \quad [\text{mark}]$$
$$\therefore 25y^2 = 16y^2 - 72y + 81$$

$$\therefore 9y^2 + 72y - 81 = 0 \quad \therefore 9(y+9)(y-1) = 0$$

$\therefore y_1 = 1$ (from diagram) [1 mark]
 $\Rightarrow$ centre of inscribed circle is $(2, 1)$
 and equation is: $(x-2)^2 + (y-1)^2 = 1$. [1 mark]